

BUS23-3

# Financial Diagnosis & Management Control

---

Albert School — 22 hours

# Contents

|                                                                        |    |
|------------------------------------------------------------------------|----|
| Preface .....                                                          | 2  |
| 1 Intermediate management balances (SIG) .....                         | 2  |
| 2 DuPont decomposition .....                                           | 4  |
| 3 Return on Invested Capital (ROIC) and ROA .....                      | 5  |
| 4 Advanced ratios and the drivers of Working Capital Requirement ..... | 6  |
| 5 Free cash flow: the number valuation uses .....                      | 8  |
| 6 Accounting adjustments for comparability .....                       | 9  |
| 7 Peer group analysis .....                                            | 10 |
| 8 Company lifecycle and financial profiles .....                       | 11 |
| 9 Budgeting and variance analysis .....                                | 12 |
| 10 Control-lever design and comprehensive diagnosis .....              | 14 |
| 10.1 Capstone: a comprehensive financial diagnosis .....               | 14 |

## Preface

In **Financial Accounting & Statements** (BUS13-3) you learned to read the three financial statements as constructed documents: to separate operating from financing, to compute EBITDA, EBIT and net income, to calculate the four families of ratios and the Working Capital Requirement (BFR), and to tell profit from cash. That course gave you **literacy** — the ability to find the numbers that matter and say what each one is.

This course turns literacy into **diagnosis** and then into **control**. Reading that a company's ROE is 13% is not yet an answer; the question a manager or investor actually asks is **why** it is 13%, **whether** that 13% is durable or fragile, and **what lever** would move it. So we decompose: profitability into its chain of intermediate balances (SIG), return into its drivers (DuPont), operating performance into a figure independent of financing (ROIC), and the operating cycle into the days that lock up cash. We compute the number valuation actually discounts — **free cash flow** — and we learn to adjust two companies' accounts so a comparison is honest. Then we close the loop the other way: budgets, variance analysis, and the design of control levers that turn a retrospective reading into forward-looking piloting.

This is the analytical engine of the Finance arc. The ROIC → free cash flow → diagnosis framework built here feeds directly into **Corporate Finance** (S3), **Valuation** (S4) and **Corporate Finance Leadership** (S6).

A word on method, continued from BUS13-3: every idea arrives as a concrete example first and the general principle second. We keep the same running company, **Lumio**, a mid-size lighting manufacturer, so that every new tool refers back to one coherent set of numbers you already half-know. Lumio's headline figures (€m): revenue 600, COGS 360, EBITDA 120, EBIT 80, net income 45; total assets 750, equity 350, financial debt 260, cash 40; inventory 120, receivables 90, payables

80. A leveraged competitor, **Helios**, and a distressed peer, **Penumbra**, join

later to sharpen the comparisons.

## 1 Intermediate management balances (SIG)

Lumio's income statement ends at €45 m of net income. But "net income fell 10%" tells a manager nothing about **where** in the chain from revenue to bottom line the erosion happened — was it pricing, labour cost, depreciation, interest, or tax? The **intermediate management**

**balances** (*soldes intermédiaires de gestion*, SIG) cut the income statement into a cascade of successive results, so that a change can be traced to the exact layer that caused it.

**Worked example — reading Lumio's profitability chain.** Walk down Lumio's figures (€m). Sales of 600 less the cost of the goods consumed to make them gives the wealth Lumio itself adds. From that we strip personnel and operating taxes to reach the cash-like operating surplus (EBE  $\approx$  EBITDA = 120); then depreciation (40) to reach operating profit (EBIT = 80); then the net cost of debt to reach pre-tax profit; then tax to reach the €45 m owners keep. Each subtraction answers a different “how well?” question.

**Definition 1** The **SIG cascade**, computed in order (French presentation; the right-hand column names the international near-equivalent):

- **Commercial margin / production of the period** — sales less the cost of goods sold or the cost of production consumed.
- **Value added** (VA) = production — purchases of goods and external services consumed. The wealth the firm itself creates, before paying labour or capital.
- **Gross operating surplus** (EBE) = VA + operating subsidies — personnel costs — taxes and duties. The cash operating result before depreciation and financing — close to **EBITDA**.
- **Operating result** (EBIT) = EBE — depreciation, amortisation and net operating provisions + other operating income.
- **Current result before tax** = operating result + financial income — financial expenses. Operations **and** financing, before exceptional items and tax.
- **Net result** = current result + exceptional result — income tax — employee profit-sharing.

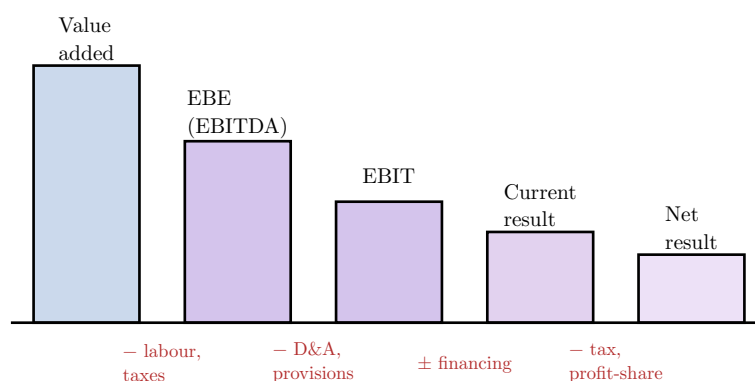


Figure 1: The SIG cascade as a waterfall. Each step removes one layer of cost — labour and operating taxes, then depreciation and provisions, then financing, then tax — so a year-on-year fall can be attributed to the precise balance where it occurs. Heights illustrate Lumio's figures.

What each balance **reveals**: value added measures the wealth Lumio itself creates (and is the base from which labour and the State are paid); EBE measures the profitability of operations before depreciation and financing choices, so it is comparable across firms with different asset ages and debt loads; EBIT adds the cost of using up fixed assets; the current result folds in the financing structure; the net result is what is finally attributable to owners.

**Common pitfall** Do not confuse **value added** with **gross profit / commercial margin**. Value added subtracts **all** external purchases (components, energy, subcontracting, rent) but **not** personnel — labour is paid out of value added, not before it. A firm that outsources

heavily shows low value added with high external charges; a vertically integrated firm shows the opposite for identical economics. Compare the **structure** of the cascade, not just the bottom line.

## 2 DuPont decomposition

Lumio earns an ROE of  $45/350 \approx 12.9\%$ . Helios, its competitor, reports an ROE of **18%** — apparently the better business. Before you believe that, ask **how** each return is built. DuPont decomposition splits ROE into three multiplicative drivers and very often overturns the naïve ranking.

**Definition 2 DuPont decomposition** writes return on equity as a product of three drivers:

$$\text{ROE} = \underbrace{\frac{\text{Net income}}{\text{Revenue}}}_{\text{net margin}} \times \underbrace{\frac{\text{Revenue}}{\text{Total assets}}}_{\text{asset turnover}} \times \underbrace{\frac{\text{Total assets}}{\text{Equity}}}_{\text{leverage}}.$$

The first two are **operating** drivers (how profitably and how intensively the firm uses its assets); the third is a **financing** driver (how much of those assets is funded by debt rather than equity).

**Worked example — same ROE gap, opposite quality.**

| Firm   | Net margin | Asset turnover | Leverage | ROE   |
|--------|------------|----------------|----------|-------|
| Lumio  | 7.5%       | 0.80           | 2.14     | 12.9% |
| Helios | 4.0%       | 0.75           | 6.00     | 18.0% |

Helios's higher ROE is **entirely** manufactured by leverage: its assets are six times its equity. On the operating drivers that Helios actually controls — margin and turnover — it is **weaker** than Lumio. Strip the leverage and Helios is the worse business; the 18% is a risk story, not a performance story.

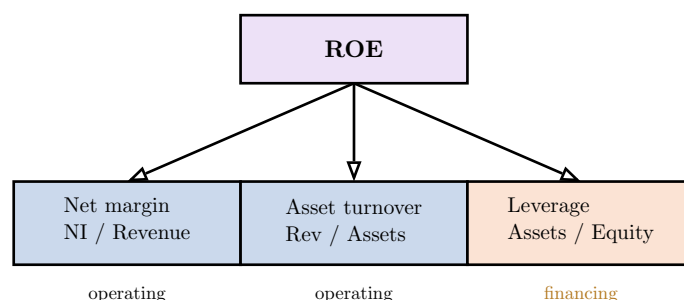


Figure 2: The DuPont tree. ROE is the product of two operating drivers and one financing driver. A return resting on the right-hand branch (leverage) is structurally more fragile than one resting on the left.

**Definition 3** By the source of the return:

- a **margin-driven** return rests on pricing power or cost control (high net margin);

- a **turnover-driven** return rests on asset efficiency (much revenue per euro of assets — the supermarket model);
- a **leverage-driven** return rests on debt (a high assets-to-equity ratio).

The strategic reading: margin- and turnover-driven returns reflect **operating** performance and tend to be **durable**. Leverage-driven returns are **fragile** — leverage is a multiplier that amplifies losses exactly as it amplifies gains, and it depends on continued access to debt at tolerable rates. When debt markets tighten or a downturn hits, the highly levered firm's ROE collapses fastest.

**Common pitfall** A rising ROE is **good news only if the operating drivers rose**. An ROE pushed up purely by taking on more debt (or by buying back shares to shrink the equity denominator) looks like improving performance but is increasing risk. Always decompose before you congratulate — and watch for ROE “improvements” that are really leverage in disguise.

### 3 Return on Invested Capital (ROIC) and ROA

DuPont exposed the problem; ROIC solves it. We want a single number that measures **operating** performance **cleansed of financing**, so that Lumio and Helios can be compared on the merits of their businesses alone. ROE cannot do this — its leverage term guarantees that capital structure leaks in. ROIC can.

**Definition 4** Return on Invested Capital:

$$\text{ROIC} = \frac{\text{NOPAT}}{\text{Invested capital}}, \quad \text{NOPAT} = \text{EBIT} \times (1 - t),$$

where  $t$  is the effective tax rate. **NOPAT** (net operating profit after tax) is what the firm would earn after tax if it had **no debt** — operating profit, taxed, but before any interest. **Invested capital** is the operating capital that funds the business:

Invested capital = Equity + net financial debt = net fixed assets + operating working capital.

The two expressions are equal: both count every euro tied up in operations, whether financed by owners or lenders.

**Return on Assets:**  $\text{ROA} = \frac{\text{Net income}}{\text{Total assets}}$  — a simpler, cruder cousin that still mixes in financing through net income.

**Worked example — Lumio's ROIC, and why it beats ROE for comparison.**  
Lumio: EBIT 80, tax rate 25%, so  $\text{NOPAT} = 80 \times 0.75 = 60$ . Invested capital = equity 350 + net debt  $(260 - 40) = 570$ . Hence

$$\text{ROIC} = 60/570 \approx 10.5\%, \quad \text{ROA} = 45/750 = 6.0\%.$$

Now imagine a twin of Lumio with **identical operations** but financed entirely by equity. Its EBIT, NOPAT and invested capital are unchanged, so its ROIC is **still 10.5%** — yet its ROE would differ from Lumio's because the equity base and interest cost differ. ROIC sees through the financing; ROE does not. That is why, when comparing firms with different capital structures, ROIC is the more reliable measure of operating performance.

The deeper purpose of ROIC is to test for **value creation**. A firm creates value only when its operations earn more than the capital they consume costs.

**Definition 5** Let **WACC** be the weighted average cost of capital — the blended after-tax cost of the firm’s debt and equity (developed fully in Corporate Finance, S3). The firm **creates** value when  $ROIC > WACC$  and **destroys** it when  $ROIC < WACC$ . The gap  $ROIC - WACC$  is the **economic spread**; multiplied by invested capital it is (approximately) the **economic profit** (EVA).

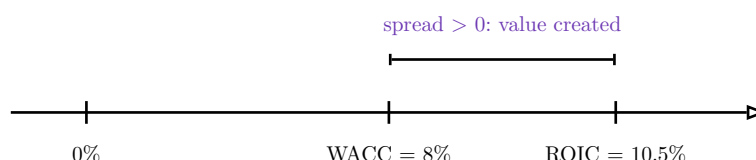


Figure 3: Value creation is a comparison, not a level. Lumio’s 10.5% ROIC exceeds an assumed 8% WACC, so the operating spread is positive and each euro of invested capital adds value. A 10.5% ROIC against a 12% WACC would destroy value despite looking healthy in isolation.

**Common pitfall** A common slip is to compute NOPAT with the **wrong** profit line. Use **EBIT** (operating profit), never net income or EBT — those already have interest subtracted, which would smuggle financing back into a measure built precisely to exclude it. Equally, keep **cash** and other non-operating assets out of invested capital; ROIC should reflect the capital that **operates**, not idle balances.

## 4 Advanced ratios and the drivers of Working Capital Requirement

You computed BFR in BUS13-3 as a single number (Lumio: €130 m). A diagnosis needs more: **where** the cash is trapped, **how many days** it stays trapped, and **how much** a given improvement would release. We convert each component of working capital into a duration, because days are comparable across firms of any size and translate directly into cash.

**Worked example — the same euro of BFR, two very different stories.** Two firms both tie up €130 m in working capital. In the first, it is mostly **inventory** sitting 120 days in a warehouse; in the second, it is mostly **receivables** — customers taking 100 days to pay. The euro figure is identical but the remedy is not: the first needs production and stocking discipline, the second needs credit and collections discipline. Days tell them apart; the total does not.

**Definition 6** The **WCR drivers**, expressed in days (use a 365-day year):

$$DIO = \frac{\text{inventory}}{\text{COGS}} \times 365, \quad DSO = \frac{\text{receivables}}{\text{revenue}} \times 365,$$

$$DPO = \frac{\text{payables}}{\text{purchases}} \times 365.$$

**DIO** (days inventory outstanding) is how long stock sits; **DSO** (days sales outstanding) is how long customers take to pay; **DPO** (days payable outstanding) is how long the firm makes its own suppliers wait. The **cash conversion cycle**

$$\text{CCC} = \text{DIO} + \text{DSO} - \text{DPO}$$

is the number of days, net, that the operating cycle locks up cash.

**Worked example — Lumio's cash conversion cycle.** With COGS and purchases  $\approx$  €360 m and revenue €600 m:

$$\text{DIO} = 120/360 \times 365 \approx 122 \text{ d}, \quad \text{DSO} = 90/600 \times 365 \approx 55 \text{ d}, \quad \text{DPO} = 80/360 \times 365 \approx 81 \text{ d}.$$

$$\text{CCC} \approx 122 + 55 - 81 = 96 \text{ days}.$$

Lumio finances about 96 days of operations out of its own pocket before a euro of margin is even counted.

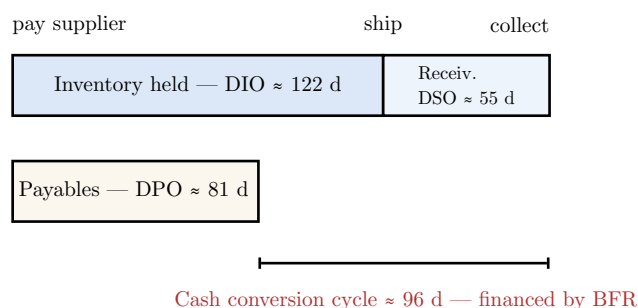


Figure 4: Lumio's operating cycle in days. Suppliers finance the first 81 days; the remaining 96 days of the inventory-plus-receivables stretch are financed by the firm itself. Shortening any segment releases cash.

Alongside the cycle, **solvency under operations** is checked by interest coverage:

**Definition 7** **Interest coverage**  $= \frac{\text{EBIT}}{\text{interest expense}}$  — how many times operating profit covers the interest due. Lumio:  $80/20 = 4.0 \times$ . Below roughly  $2 \times$ , a firm has little cushion: a modest dip in EBIT puts debt service at risk.

**Modelling the cash impact.** Each day removed from a driver releases cash equal to one day of that driver's base. Cutting DSO by 10 days releases about  $10 \times \text{revenue}/365 = 10 \times 600/365 \approx$  €16.4 m of cash — a one-time inflow that permanently lowers the capital the business must carry.

**Common pitfall** Every working-capital improvement has a **trade-off**, and optimising blindly does real damage:

- cutting **inventory days** too far risks stock-outs and lost sales;
- cutting **receivable days** by tightening credit terms can drive customers to competitors;
- stretching **payable days** strains supplier relationships and forfeits early-payment discounts (often an expensive implicit interest rate).

The goal is not the lowest possible CCC but the **cheapest sustainable** one.

## 5 Free cash flow: the number valuation uses

Lumio earned €45 m of net income. How much cash was actually **free** — available to hand to the people who financed the firm, after paying for everything the business needed to keep running and growing? That figure, not net income, is what every valuation discounts. Net income is an accounting opinion; free cash flow is the cash that can be distributed.

**Definition 8 Free cash flow to the firm** — cash available to **all** providers of capital (debt and equity), before financing flows:

$$\text{FCFF} = \text{NOPAT} + \text{D\&A} - \text{capex} - \Delta\text{WCR}.$$

**Free cash flow to equity** — what remains for **shareholders** after debt is serviced and new debt raised:

$$\text{FCFE} = \text{FCFF} - \text{interest} \times (1 - t) + \text{net borrowing}.$$

We build FCFF from NOPAT (not net income) precisely so that it is independent of financing — the same reason ROIC used NOPAT. We add back **D&A** because it reduced profit without any cash leaving; subtract **capex** because replacing and expanding the asset base is a real cash outflow the income statement ignored this period; and subtract the **increase in WCR** because every extra euro tied up in inventory and receivables is a euro of cash the business swallowed.

**Worked example — Lumio: from NOPAT to FCFF to FCFE (€m).** Take D&A = 40, capex = 60, and a €10 m rise in WCR this year; interest = 20, tax rate 25%, net new borrowing = 0.

|                                                  |           |
|--------------------------------------------------|-----------|
| NOPAT = EBIT $\times$ (1 - t) = 80 $\times$ 0.75 | 60        |
| + D&A                                            | 40        |
| – Capex                                          | (60)      |
| – Increase in WCR                                | (10)      |
| <b>FCFF</b>                                      | <b>30</b> |
| – After-tax interest = 20 $\times$ 0.75          | (15)      |
| + Net borrowing                                  | 0         |
| <b>FCFE</b>                                      | <b>15</b> |

Lumio turned €45 m of accounting profit into just €30 m of cash free to all investors, and €15 m free to shareholders — the rest was consumed by capex and by growth in working capital. The gap between €45 m and €15 m is exactly what a dividend-focused shareholder must understand.

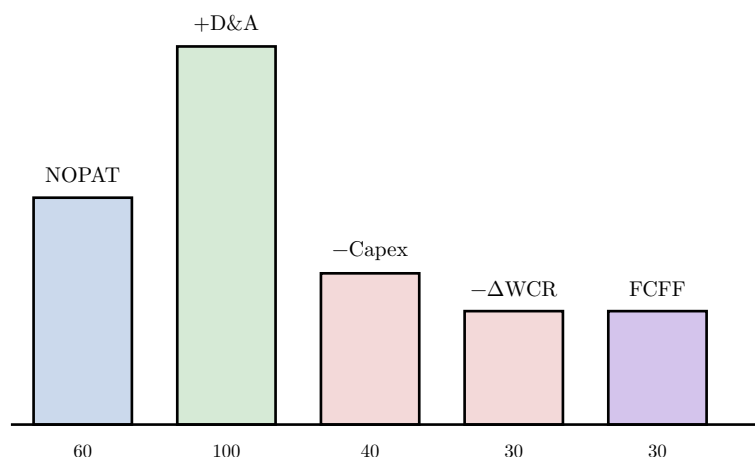


Figure 5: The FCFF bridge for Lumio (€m). Starting from NOPAT, depreciation is added back (non-cash), then capex and the increase in working capital are removed (real cash uses), leaving €30 m of free cash flow to the firm.

**Common pitfall** Three recurring errors:

1. Building FCFF from **net income** instead of NOPAT, then forgetting to add back after-tax interest — this double-counts the financing effect.
2. Omitting  $\Delta WCR$ . For a fast-growing firm this term is large and negative for cash; ignoring it makes a cash-hungry growth company look self-funding.
3. Confusing FCFF and FCFE. FCFF is **before** debt service and is discounted at the WACC; FCFE is **after** and is discounted at the cost of equity. Mixing the cash flow of one with the discount rate of the other is the classic valuation blunder.

## 6 Accounting adjustments for comparability

We are about to compare Lumio with peers. But reported financials embed **accounting choices**, and two firms with identical economics can look very different on paper merely because they chose differently. Before any honest comparison, we normalise those choices away.

**Worked example — two identical firms, three accounting forks.** Lumio and a rival run the same business, but: Lumio **expenses** its R&D while the rival **capitalises** it; Lumio **owns** its warehouses while the rival **leases** them off balance sheet; Lumio pays staff in cash while the rival pays partly in **stock**. Result: the rival reports higher operating profit, more assets, and lower cash wages — better-looking margins and turnover — for **no real difference in performance**. Every gap is an accounting artefact. Adjustment removes it.

**Definition 9** The three core **comparability adjustments**:

- **R&D capitalisation.** Treat R&D as an investment: add this year's R&D back to operating profit, create an R&D asset on the balance sheet, and amortise it over its useful life. This puts an R&D-expensing firm on the same footing as an R&D-capitalising one (and lifts the expenser's invested capital, lowering its flattering ROIC toward truth).
- **Operating-lease capitalisation.** Recognise a **right-of-use asset** and a matching **lease liability**; replace the rent expense with depreciation plus implied interest. This brings an off-balance-sheet lease onto the books so a lessee is comparable to an owner — and reveals the lease's hidden debt.

- **Stock-based-compensation normalisation.** Treat SBC as the real operating expense it is (employees were paid, just in shares), so it is not quietly excluded from “adjusted” profit; and adjust the share count for dilution.

**Common pitfall** Unadjusted comparisons do not just add noise — they **bias** in a predictable direction. Off-balance-sheet leases understate both assets and debt, flattering ROIC **and** leverage at once. Capitalised R&D inflates current profit relative to an expenser. “Adjusted EBITDA” that addsback SBC pretends a real cost is free. Whenever a company’s own presentation makes it look better than a plain reading would, ask which choice is doing the work — and undo it before you compare.

## 7 Peer group analysis

A ratio means nothing in isolation (BUS13-3): Lumio’s 10.5% ROIC is good or bad only **relative** to something. The right something is a **peer group** — but a careless peer group produces confident nonsense. The craft is in selecting comparables and then **controlling** for the ways they still differ.

**Definition 10** A **peer group** is a set of companies comparable enough to the subject that ratio differences can be read as performance differences. Selection runs:

1. **Select** firms in the same business model (what they sell, to whom, how they earn).
2. **Compute** the same ratios, on the same (adjusted) basis, across the group.
3. **Control** for residual differences in size, geography, capital structure and accounting regime.
4. **Interpret** the remaining gaps.

**Worked example — is Lumio's margin gap real?.** Lumio’s EBIT margin is 13.3%; a peer’s is 19%. Before concluding Lumio is worse: is the peer **larger** (scale economies in purchasing)? In a **different geography** (higher prices, lower labour cost)? Reporting under a different **accounting regime** (capitalising costs Lumio expenses)? A peer twice the size, selling in a premium market, capitalising its R&D, will out-margin Lumio for reasons that have nothing to do with management quality. Control for those, and the “6-point gap” may shrink to one.

**Definition 11** After controlling, classify each gap:

- a **genuine performance** gap survives when peers are truly comparable and the difference persists after controls — it points to real operating strength or weakness, and is **actionable**;
- a **structural** gap is explained away by size, geography, accounting regime or business model — it reflects **what the company is**, not **how well it is run**, and chasing it wastes effort.

**Common pitfall** The two failure modes are opposite and equally common. **Too-narrow** peer groups (three firms, all friends of the analyst’s thesis) manufacture whatever conclusion

you want. **Too-broad** groups (“the industry”) lump together business models so different that the median is meaningless. And never compare a ratio across peers without first applying the adjustments of the previous chapter — otherwise you are ranking accounting choices, not companies.

## 8 Company lifecycle and financial profiles

The same diagnostic signal means opposite things at different ages. Negative free cash flow is a **death rattle** in a mature firm and a **healthy sign of ambition** in a young one. Before judging any number, locate the company on its lifecycle, then set expectations accordingly.

**Definition 12** Typical financial profile by **lifecycle stage**:

| Stage    | Margins / growth                           | Free cash flow                            | Financing                |
|----------|--------------------------------------------|-------------------------------------------|--------------------------|
| Startup  | negative or thin margins, very high growth | strongly negative                         | external equity          |
| Growth   | improving margins, high growth             | often still negative (reinvestment + WCR) | equity + debt            |
| Maturity | stable margins, low growth                 | positive, substantial                     | self-funds; returns cash |
| Decline  | falling margins and revenue                | positive but shrinking; asset sales       | repays / returns         |

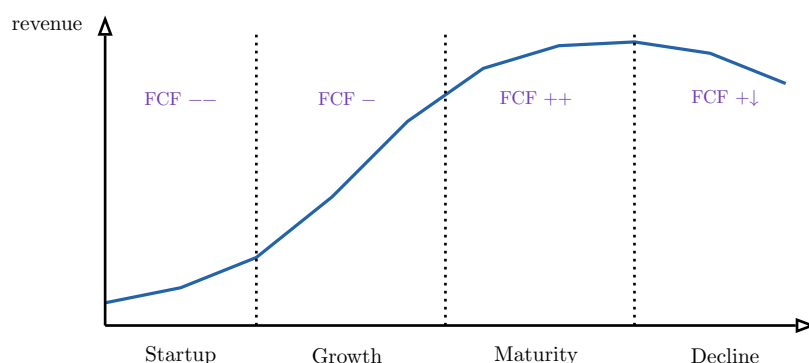


Figure 6: Revenue over the lifecycle, with the typical free-cash-flow signature annotated. The diagnostic meaning of “negative FCF” inverts between the growth and mature stages — read every signal against the stage, never absolutely.

**Worked example — two firms, identical FCF of –€20 m.** A SaaS startup growing 80% a year burns €20 m: normal — it is investing ahead of a large future. A mature lighting maker the size of Lumio, growing 1%, burns €20 m: alarming — its operations should be throwing off cash, so something (collapsing margins, ballooning WCR, an unaffordable dividend) is broken. Same number; opposite diagnosis.

**Common pitfall** The error is **applying a single threshold across stages**. “FCF must be positive”, “growth above 20% is good”, “margins below 5% are weak” are all stage-dependent. Calibrate every alert threshold (next chapter) to the company’s stage, and re-calibrate as it moves between stages — the dangerous moment is the transition, when growth-stage habits (heavy reinvestment, cash burn) persist into a maturity that no longer justifies them.

## 9 Budgeting and variance analysis

So far we have **diagnosed the past**. Management control also **steers the future**: it sets a plan (the budget), measures what actually happened against it, and decomposes the gap (the variance) into causes a manager can act on. The art of variance analysis is to split one aggregate miss into volume, price and mix — because each points to a different owner and a different remedy.

**Definition 13** A **driver-based budget** projects results from operational drivers — volume, price, unit cost — rather than by nudging last year’s figures up a few percent. A **fixed annual budget** is set once for the year; a **rolling forecast** is re-extended each period to keep a constant horizon (e.g. always 12 months ahead).

**Worked example — Lumio's lamp division misses its profit target.** Budget: sell 100k lamps at €50, unit cost €30 → budgeted margin €20/unit = €2.0 m. Actual: sold 110k lamps at €48, unit cost €31 → actual margin €17/unit = €1.87 m. The €0.13 m shortfall hides three offsetting moves: **more** units (good), a **lower** price (bad), a **higher** cost (bad). One number, three stories — which is exactly what decomposition recovers.

**Definition 14** **Variance decomposition** against budget (holding the others at budget when isolating each):

- **Volume variance** = (actual qty – budget qty) × budgeted unit margin.
- **Price variance** = (actual price – budget price) × actual qty.
- **Mix variance** = the effect of a shift in the **proportion** of products sold, holding total volume constant.
- **Cost (standard-vs-actual) variance** = (standard cost – actual cost) × actual qty.

Favourable variances raise profit; unfavourable ones lower it.

#### Worked example — decomposing the €0.13 m miss.

- Volume:  $(110 - 100)\text{k} \times \text{€}20 = +\text{€}200\text{k}$  (favourable).
- Price:  $(48 - 50) \times 110\text{k} = -\text{€}220\text{k}$  (unfavourable).
- Cost:  $(30 - 31) \times 110\text{k} = -\text{€}110\text{k}$  (unfavourable).

Net:  $+200 - 220 - 110 = -\text{€}130\text{k}$ , reconciling exactly. The division did not “underperform” uniformly — it **won on volume and lost on price and cost**. The manager now knows to defend price and investigate the cost overrun, not to push for yet more volume.

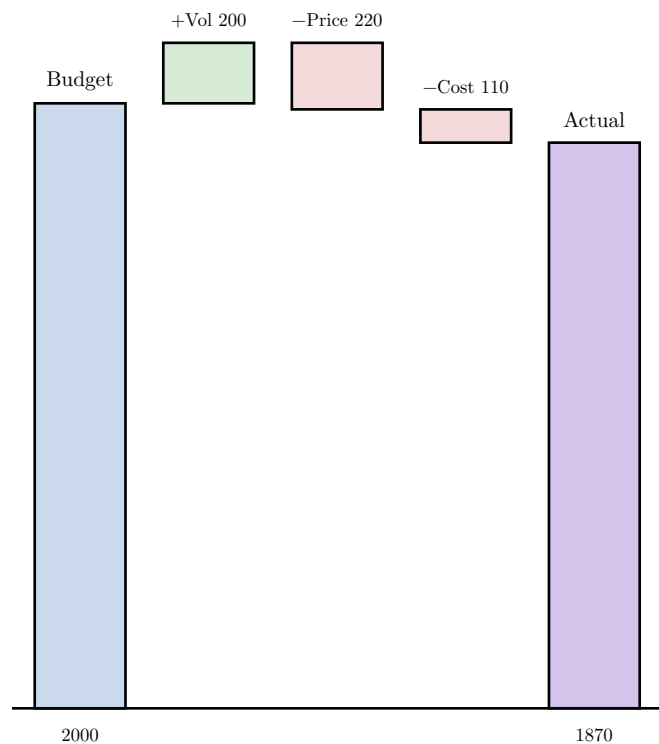


Figure 7: The variance bridge (€k). The €130k gap between budgeted and actual margin decomposes into a favourable volume effect and unfavourable price and cost effects — each attributable to a different decision and owner.

**Definition 15** By controllability, a variance is:

- **actionable / operational** when it arises from execution within the unit’s control (waste, a fixable price concession, a one-off cost overrun) — correct it operationally;
- **structural** when it reflects a durable shift in conditions (a permanent rise in input prices, a demand collapse) — no amount of operational effort fixes it; it requires **strategic redesign**.

**Common pitfall** Do not “manage” a structural variance operationally — punishing a plant manager for an industry-wide raw-material spike wastes effort and credibility. And beware the opposite: labelling a genuinely operational failure “structural” to excuse it. The diagnostic question is whether the cause is **inside or outside** the unit’s control and **temporary or durable**.

## 10 Control-lever design and comprehensive diagnosis

Diagnosis tells you what is wrong; **control-lever design** builds the instrument panel that keeps it from going wrong again. A business unit cannot be steered by the full annual accounts — those arrive too late and contain too much. It needs a **small** set of leading indicators, watched at the **right frequency**, with **thresholds** that trigger action before a problem becomes a crisis.

**Definition 16** A **control-lever system** for a business unit specifies: the **piloting indicators** to monitor (a handful, each tied to a genuine value driver), the **monitoring frequency** for each, and the **alert thresholds** at which a deviation demands intervention.

The design choices follow from the diagnosis:

- **Which indicators?** Choose the few that drive the unit’s value and that the unit can actually move — for Lumio’s lamp division, perhaps order intake, gross margin per unit, DSO, and on-time delivery. Avoid vanity metrics and avoid drowning the pilot in dials.
- **How often?** Match the cadence to how fast the indicator moves and how fast action is needed: cash and order intake daily or weekly; margins and DSO monthly; strategic ratios quarterly. Watching a slow indicator daily wastes attention; watching a fast one quarterly misses the fire.
- **What threshold?** Set the alert at the deviation beyond which a manager must act — wide enough to ignore noise, tight enough to catch a real drift early. Calibrate to the lifecycle stage (previous chapter): a growth unit tolerates wider swings than a mature one.

**Definition 17** **Rolling forecast vs fixed annual budget.** A **rolling forecast** outperforms when operating conditions are **volatile or fast-changing**, so a once-a-year target goes stale and demotivates; it costs more effort and can encourage short-termism. A **fixed annual budget** suffices — and is cheaper and more stable — when conditions are **predictable**. Match the instrument to the environment, not to fashion.

**Worked example — two units, two regimes.** Lumio’s mature **components** unit sells long-cycle parts to stable industrial customers: a fixed annual budget, reviewed monthly, fits. Its new **smart-lighting** unit faces fast-moving competitors and volatile demand: a quarterly rolling forecast keeps its targets honest. Same company, two control regimes — chosen by the volatility of each unit’s world, not imposed uniformly.

### 10.1 Capstone: a comprehensive financial diagnosis

Everything converges here: a single, remediation-oriented diagnosis of a real, possibly distressed company. Meet **Penumbra**, a lighting competitor in trouble. We run the whole toolkit in sequence, each tool narrowing the question the next one answers.

**Definition 18** The **diagnostic sequence**:

1. **SIG** — walk the profitability chain to locate **where** it breaks (is the loss at value added? at EBE? only after financing?).
2. **DuPont** — identify the **failing return driver**: margin, turnover, or leverage.
3. **ROIC vs WACC** — judge whether operations create or destroy value, independent of financing.
4. **WCR drivers** — find cash trapped in the operating cycle (which days are bloated?).
5. **Free cash flow** — assess true cash generation and solvency.
6. **Peer comparison**, on an **adjusted** basis — separate genuine underperformance from structural difference.

Then write up: name the failing driver(s), assess **severity**, and propose **quantified** remediation targeted at the diagnosed cause.

**Worked example — diagnosing Penumbra. SIG.** Value added is healthy, but EBE has collapsed: personnel costs have grown far faster than sales. The break is at the **labour** layer, not pricing.

**DuPont.** ROE is 9% but built on leverage 4.5×; net margin is only 2%. The return is **fragile and margin-poor** — a leverage story masking weak operations.

**ROIC.** ROIC is 5% against an estimated WACC of 9%: a **negative spread** — Penumbra is **destroying** value with every euro invested.

**WCR.** DSO has ballooned to 95 days (lax collections); €30 m of cash is trapped that a return to Lumio's 55-day norm would release.

**FCF.** FCFF is barely positive and FCFE negative — the dividend is being funded by **new borrowing**, which is unsustainable.

**Peers (adjusted).** Even after capitalising Penumbra's heavy operating leases — which had hidden much of its true debt — it trails the peer median on margin and ROIC. The gap is **genuine**, not structural.

**Diagnosis & remediation.** The failing drivers are **operating margin** (labour cost out of control) and **working-capital discipline** (DSO). Severity: **high** — value destruction plus debt-funded dividends. Remediation, quantified: (1) cut the labour-cost ratio back toward the peer norm to restore EBE (target +4 pts of EBIT margin); (2) bring DSO from 95 to 60 days, releasing ≈ €25 m of cash; (3) suspend the dividend until FCFE turns positive; (4) use the released cash to cut the leverage that makes the whole structure fragile.

**Common pitfall** A diagnosis that stops at “ROE is low” is useless. The value is in the **chain**: ROE is low **because** the net margin is low (DuPont) **because** labour cost has outrun sales (SIG) — and the fix therefore targets **labour**, not, say, asset turnover. Always push from symptom to driver to cause to remedy, and **quantify** the remedy. An unquantified recommendation (“improve collections”) is an opinion; “cut DSO from 95 to 60 days to release €25 m” is a plan.

**The thread of this book.** BUS13-3 taught you to read the statements; this course taught you to **diagnose** and **steer**. The SIG cascade locates where profitability breaks; DuPont names the failing return driver; ROIC measures operating performance cleansed of financing and tests it against the cost of capital; the WCR drivers find trapped cash; free cash flow reveals what is truly distributable and feeds valuation; adjustments and peer analysis make comparison honest; and lifecycle awareness keeps every threshold calibrated. Turning the loop forward, driver-based budgets and variance analysis attribute the gap between plan and reality to actionable causes, and control-lever design builds the instrument panel that pilots the unit. The synthesis — SIG → DuPont → ROIC → WCR → FCF → peers → quantified remediation — is the diagnostic engine that **Corporate Finance**, **Valuation** and **Corporate Finance Leadership** all assume you can run.